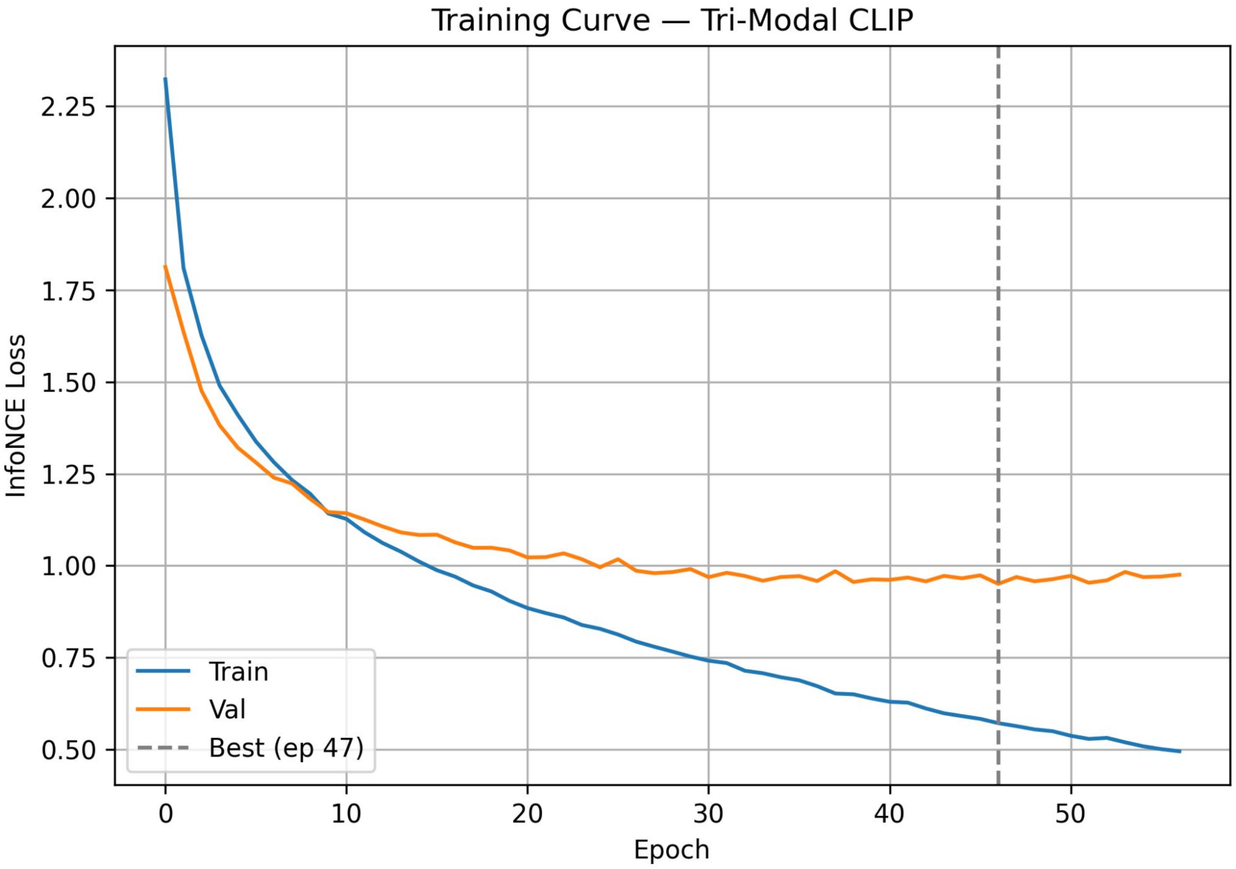
Primary Tumor and Metastasis Histology and Spatial Transcriptomics Data

Zenodo Spatial Transcriptomics Project Updates

3-Projector CLIP-style Architecture is used:

- Visium Projector downsampled to 252 dims
- Gene Projector upsampled to 252 dims
- Cell Projector upsampled to 252 dims

The model of this training curve used UNI2-h, scVI and RCTD embeddings



Zenodo PDAC Dataset Analysis

Patient-Image Mapping Summary:

1. **30 10x Genomics Visium V2 H&E Images with spatial transcriptomics data**:
   a. Patient 1 = 1 Image (IU_PDA_T1)
   b. Patient 2 = 4 Images (IU_PDA_HM2, IU_PDA_HM2_2, IU_PDA_NP2, IU_PDA_T2)
   c. Patient 3 = 2 Images (IU_PDA_HM3, IU_PDA_T3)
   d. Patient 4 = 2 Images (IU_PDA_HM4, IU_PDA_T4)
   e. Patient 5 = 1 Image (IU_PDA_HM5)
   f. Patient 6 = 3 Images (IU_PDA_HM6, IU_PDA_LNM6, IU_PDA_T6)
   g. Patient 7 = 1 Image (IU_PDA_LNM7)
   h. Patient 8 = 3 Images (IU_PDA_HM8, IU_PDA_LNM8, IU_PDA_T8)
   i. Patient 9 = 2 Images (IU_PDA_HM9, IU_PDA_T9)
   j. Patient 10 = 4 Images (IU_PDA_HM10, IU_PDA_LNM10, IU_PDA_NP10, IU_PDA_T10)
   k. Patient 11 = 3 Images (IU_PDA_HM11, IU_PDA_NP11, IU_PDA_T11)
   l. Patient 12 = 3 Images (IU_PDA_HM12, IU_PDA_LNM12, IU_PDA_T12)
   m. Patient 13 = 1 Image (IU_PDA_HM13)

Zenodo PDAC Dataset Analysis

Tissue Spot Data Summary:

Tissue Image	n_spots
IU_PDA_T1	3530
IU_PDA_HM2	2478
IU_PDA_HM2_2	959
IU_PDA_NP2	2995
IU_PDA_T2	4118
IU_PDA_HM3	1176
IU_PDA_T3	4354
IU_PDA_HM4	1841
IU_PDA_T4	3621
IU_PDA_HM5	3038

Tissue Image	n_spots
IU_PDA_HM6	1666
IU_PDA_LNM6	3745
IU_PDA_T6	3397
IU_PDA_LNM7	3186
IU_PDA_HM8	4032
IU_PDA_LNM8	3407
IU_PDA_T8	3779
IU_PDA_HM9	4032
IU_PDA_T9	3526
IU_PDA_HM10	2348

Tissue Image	n_spots
IU_PDA_LNM10	4147
IU_PDA_NP10	2966
IU_PDA_T10	2714
IU_PDA_HM11	3931
IU_PDA_NP11	3859
IU_PDA_T11	2777
IU_PDA_HM12	2961
IU_PDA_LNM12	3213
IU_PDA_T12	3642
IU_PDA_HM13	2182

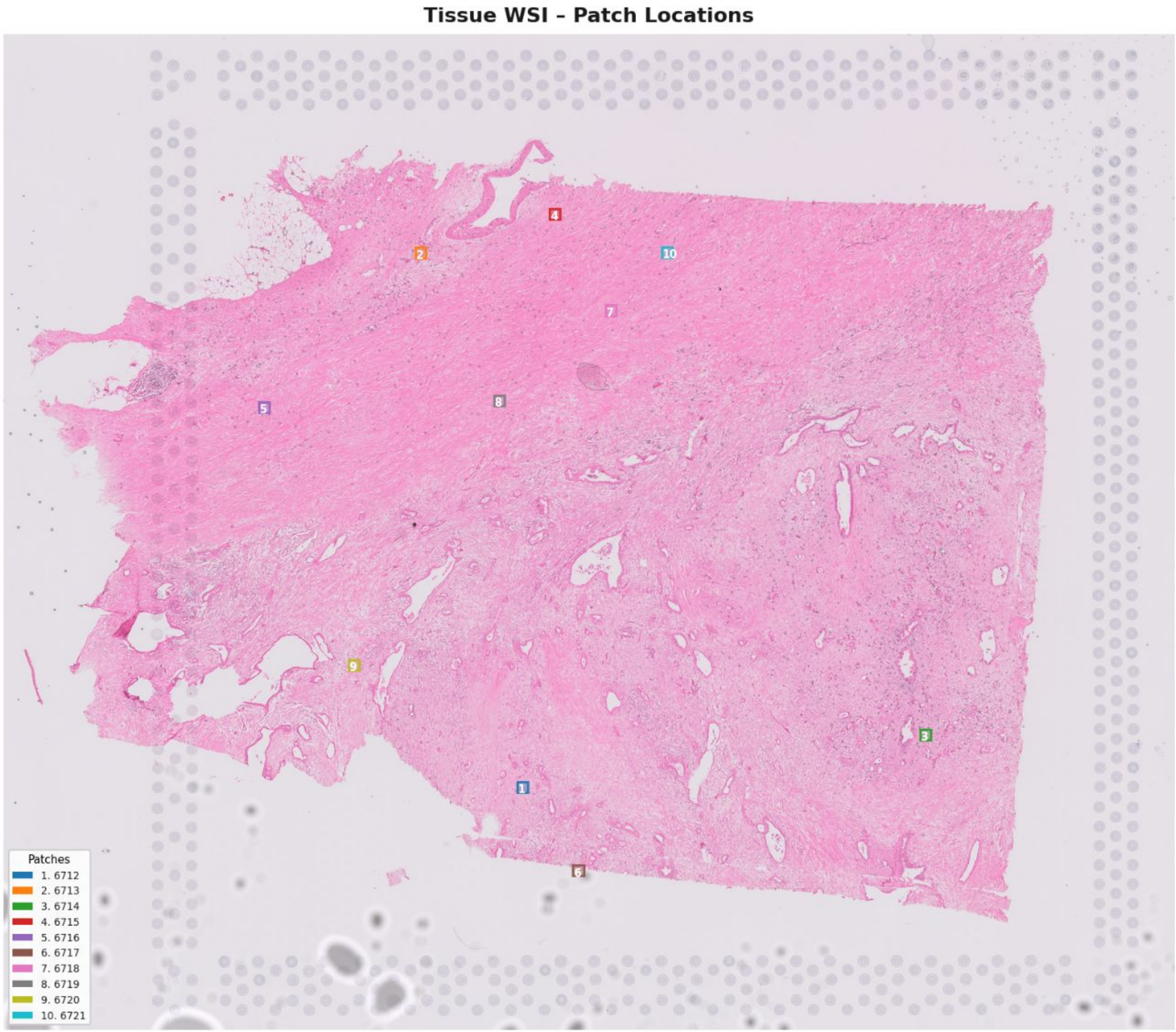


Image: IU_PDA_T1

Tumour Region Image of Patient 1

Patching Process:

- Visium V2, Spot-centered and Spot-sized with black padding patches.
- Patches extracted from .tif files
- First 10 Patches Extracted

First 10 Patches of IU_PDA_T1

1. patch_006712_PT_1
PDACP_10_PDACP_10
AAACAAGTATCTCCCA_1_50
102

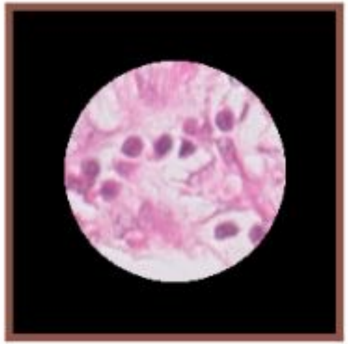
2. patch_006713_PT_1
PDACP_10_PDACP_10
AAACACCAATAACTGC_1_59
19

3. patch_006714_PT_1
PDACP_10_PDACP_10
AAACAGAGCGACTCCT_1_14
94

4. patch_006715_PT_1
PDACP_10_PDACP_10
AAACAGGGTCTATATT_1_47
13

5. patch_006716_PT_1
PDACP_10_PDACP_10
AAACAGTGTTCCTGGG_1_73
43

6. patch_006717_PT_1
PDACP_10_PDACP_10
AAACCCGAACGAAATC_1_45
115



7. patch_006718_PT_1
PDACP_10_PDACP_10
AAACCGGGTAGGTACC_1_42
28

8. patch_006719_PT_1
PDACP_10_PDACP_10
AAACCGTTCGTCCAGG_1_52
42

9. patch_006720_PT_1
PDACP_10_PDACP_10
AAACCTAAGCAGCCGG_1_65
83

10. patch_006721_PT_1
PDACP_10_PDACP_10
AAACCTCATGAAGTTG_1_37
19

Clinical Validation — Patient & Patch Metadata | T1 (Patient 1)

Table 1A — Patient Clinical Metadata (T1)	
Field	**Value**
Patient ID	PT_1
Sample ID	S13-11414_B9
Slide Name	V12M07-117
Area Code	A1
Origin	Pancreas
Age	56
Gender	Male
Ethnicity	White
Tumour Location	Body and tail
AJCC Stage	pT3pN1pM1
Histology	PDAC
Tumour Grade	G2
Neoadjuvant Tx	Yes
Chemo	YES
R0 Resection	YES
PNI	YES
LVI	YES
Nodes Collected	YES
Nodes Positive	3 OUT 7
Surgery Date	5/6/13

Clinical Validation — Patient & Patch Metadata | T1 (Patient 1)

Table 1B — Visium Scale Information (T1)	
Field	**Value**
Mean Pixel Spacing	271.383
Microns / Pixel	0.368483
Pixels / Micron	2.71383
Spot Diameter (px)	149.261
Spot Radius (px)	74.6303
Scale Spot	0.0812612
Scale Fiducial	284.093
Scale Hi-Res	0.0812612
Scale Lo-Res	0.0243784

Clinical Validation — Patient & Patch Metadata | T1 (Patient 1)

Table 2 — T1 Patch / Spot Metadata (top 10 patches)						
Patch ID	**Barcode**	**Tissue Row**	**Tissue Col**	**Image Row**	**Image Col**	**Patch Filename**
6712	PDACP_10 AAACAAGTATCTCCCA-1	50	102	15698	10783	patch_006712_PT_1 PDACP_10_PDACP_10 AAACAAGTATCTCCCA_1 50_102
6713	PDACP_10 AAACACCAATAACTGC-1	59	19	4473	8648	patch_006713_PT_1 PDACP_10_PDACP_10 AAACACCAATAACTGC_1 59_19
6714	PDACP_10 AAACAGAGCGACTCCT-1	14	94	14603	19255	patch_006714_PT_1 PDACP_10_PDACP_10 AAACAGAGCGACTCCT_1 14_94
6715	PDACP_10 AAACAGGGTCTATATT-1	47	13	3657	11471	patch_006715_PT_1 PDACP_10_PDACP_10 AAACAGGGTCTATATT_1 47_13
6716	PDACP_10 AAACAGTGTTCCTGGG-1	73	43	7724	5357	patch_006716_PT_1 PDACP_10_PDACP_10 AAACAGTGTTCCTGGG_1 73_43
6717	PDACP_10 AAACCCGAACGAAATC-1	45	115	17455	11962	patch_006717_PT_1 PDACP_10_PDACP_10 AAACCCGAACGAAATC_1 45_115
6718	PDACP_10 AAACCGGGTAGGTACC-1	42	28	5684	12651	patch_006718_PT_1 PDACP_10_PDACP_10 AAACCGGGTAGGTACC_1 42_28
6719	PDACP_10 AAACCGTTCGTCCAGG-1	52	42	7582	10300	patch_006719_PT_1 PDACP_10_PDACP_10 AAACCGTTCGTCCAGG_1 52_42
6720	PDACP_10 AAACCTAAGCAGCCGG-1	65	83	13133	7248	patch_006720_PT_1 PDACP_10_PDACP_10 AAACCTAAGCAGCCGG_1 65_83
6721	PDACP_10 AAACCTCATGAAGTTG-1	37	19	4465	13826	patch_006721_PT_1 PDACP_10_PDACP_10 AAACCTCATGAAGTTG_1 37_19

All the spot/patch metadata of the 10 patches are shown with patchID and barcode as a unique identifier

Clinical Validation — Gene Expression Analysis

Table 1 — Spot Gene Metadata (first 10 spots)								
Spot ID	Gene Count	Total Expression	Mean Expression	Max Expression	Total Counts	Genes Detected	Norm. Gene Count	% Mitochondrial
6712	990	1294	0.07232	13	1294	990	0.7651	0
6713	424	543	0.03035	19	543	424	0.7808	0
6714	995	1240	0.0693	14	1240	995	0.8024	0
6715	324	378	0.02113	5	378	324	0.8571	0
6716	492	621	0.03471	12	621	492	0.7923	0
6717	1403	1899	0.1061	27	1899	1403	0.7388	0
6718	225	252	0.01408	6	252	225	0.8929	0
6719	323	389	0.02174	8	389	323	0.8303	0
6720	2413	4111	0.2298	91	4111	2413	0.587	0
6721	337	406	0.02269	8	406	337	0.83	0

All the spot gene metadata of the 10 patches are shown with spot ID as a unique identifier
Normalized Gene Count = Genes Detected / Total Counts

Clinical Validation — Marker Gene Presence Analysis

Panel A — Marker Gene Detection Heatmap (10 Spots × 6 Cell Types)

	Epithelial	Stromal	Immune	Pancreatic	Endothelial	Fibroblast
6712	1/5 EPCAM	3/3 Collagen Fibronectin Actin	0/5 —	0/5 —	0/4 —	3/4 FSP1 VIM PDGFRA
6713	0/5 —	3/3 Collagen Fibronectin Actin	0/5 —	0/5 —	0/4 —	1/4 VIM
6714	2/5 KRT8 CDH1	2/3 Collagen Actin	1/5 CD45	0/5 —	0/4 —	2/4 FSP1 VIM
6715	1/5 TJP1	2/3 Collagen Actin	0/5 —	0/5 —	0/4 —	1/4 VIM
6716	0/5 —	3/3 Collagen Fibronectin Actin	0/5 —	0/5 —	0/4 —	2/4 VIM PDGFRA
6717	3/5 EPCAM KRT8 TJP1	3/3 Collagen Fibronectin Actin	1/5 CD45	0/5 —	0/4 —	4/4 FSP1 VIM PDGFRA THY1
6718	0/5 —	3/3 Collagen Fibronectin Actin	0/5 —	0/5 —	0/4 —	1/4 VIM
6719	0/5 —	2/3 Collagen Fibronectin	0/5 —	0/5 —	0/4 —	3/4 FSP1 VIM THY1
6720	1/5 CDH1	3/3 Collagen Fibronectin Actin	0/5 —	0/5 —	2/4 PECAM1 CDH5	4/4 FSP1 VIM PDGFRA THY1
6721	0/5 —	2/3 Collagen Actin	0/5 —	0/5 —	0/4 —	1/4 FSP1

Fraction of Panel Detected: 100% (Full) · 50% · 0% (None)

Cell Value = genes found / panel total
Gene names listed = those detected

Clinical Validation — Cell Composition Analysis

Table 1 — Spot Cell-Type Composition Matrix

Cell Type	Spot 1 PDACP_10 AAACAAGTATCTCCCA-1	Spot 2 PDACP_10 AAACACCAATAACTGC-1	Spot 3 PDACP_10 AAACAGAGCGACTCCT-1	Spot 4 PDACP_10 AAACAGGGTCTATATT-1	Spot 5 PDACP_10 AAACAGTGTTCCTGGG-1	Spot 6 PDACP_10 AAACCCGAACGAAATC-1	Spot 7 PDACP_10 AAACCGGGTAGGTACC-1	Spot 8 PDACP_10 AAACCGTTCGTCCAGG-1	Spot 9 PDACP_10 AAACCTAAGCAGCCGG-1	Spot 10 PDACP_10 AAACCTCATGAAGTTG-1
B cells	0.0581	0.2030	0.0040	0.0414	0.0754	0.0143	3.17e-05	0.1163	0.0030	0.0394
C1Q-TAM	0.0689	2.97e-05	0.0140	0.0774	0.0389	0.0343	0.0619	0.0465	0.2029	2.94e-05
CD4+ cells	0.0172	2.97e-05	0.1258	2.10e-05	3.00e-05	0.0888	3.17e-05	2.10e-05	2.73e-05	0.0172
CD8-NK cells	1.30e-05	0.0012	5.85e-05	2.10e-05	3.00e-05	1.99e-05	3.17e-05	2.10e-05	2.73e-05	2.94e-05
DCs	0.0368	2.97e-05	0.0453	2.10e-05	0.1153	0.0717	0.0523	0.0580	0.0167	2.94e-05
Endothelial cells	0.0033	0.0106	5.90e-05	0.0234	0.0062	0.0091	3.17e-05	2.10e-05	0.0076	2.94e-05
FCN1-TAM	0.0709	2.97e-05	5.85e-05	0.0739	3.00e-05	1.99e-05	0.1669	2.10e-05	2.73e-05	0.0620
Hepatocytes	3.82e-04	2.97e-05	0.0024	2.10e-05	3.00e-05	1.99e-05	3.17e-05	2.10e-05	5.60e-04	2.94e-05
iCAF	0.0380	0.1013	0.2372	0.1513	0.1571	0.1243	0.0088	0.0820	6.04e-05	0.2985
myCAF	0.4930	0.5378	0.2095	0.6264	0.6068	0.4958	0.5318	0.6367	0.6682	0.5503
Normal Epithelial cells	1.30e-05	2.97e-05	5.85e-05	2.10e-05	3.00e-05	1.99e-05	3.17e-05	2.10e-05	2.73e-05	2.94e-05
Proliferative T cells	1.30e-05	0.0450	5.85e-05	5.13e-04	3.00e-05	1.99e-05	3.17e-05	0.0420	0.0129	2.94e-05
PVL	0.0404	0.0129	0.0092	1.68e-04	6.33e-05	0.0439	3.58e-05	3.15e-05	4.75e-05	1.67e-04
SPP1-TAM	1.30e-05	0.0534	5.85e-05	2.10e-05	3.00e-05	1.99e-05	0.1401	2.10e-05	2.73e-05	0.0323
Tumor Epithelial cells	0.1729	0.0347	0.3523	4.80e-04	3.00e-05	0.1177	0.0379	0.0183	0.0879	2.94e-05

High

Mid

Low

Cell-Type Composition (normalization = Cell type proportion / All cell type proportions of that spot) of 15 Cell Types shown for the 10 Spots

Clinical Validation — Marker Gene Presence Analysis

Panel B — Marker Gene Reference Dictionary (for Pathologist Review)

Epithelial	Stromal	Immune	Pancreatic	Endothelial	Fibroblast
EPCAM	Collagen (COL1A1)	CD45 (PTPRC)	PRSS1	PECAM1	FSP1 (S100A4)
KRT8		CD3E	PRSS2	CDH5	VIM
KRT19	Fibronectin (FN1)	CD8A	PNLIP	KDR	PDGFRA
CDH1		CD4	CTRC	GATA2	THY1
TJP1	Actin (ACTA2)	CD19	CEL		

This is the Marker Gene Reference Dictionary where key marker genes have been used to identify the tissue regions in IU_PDA_T1

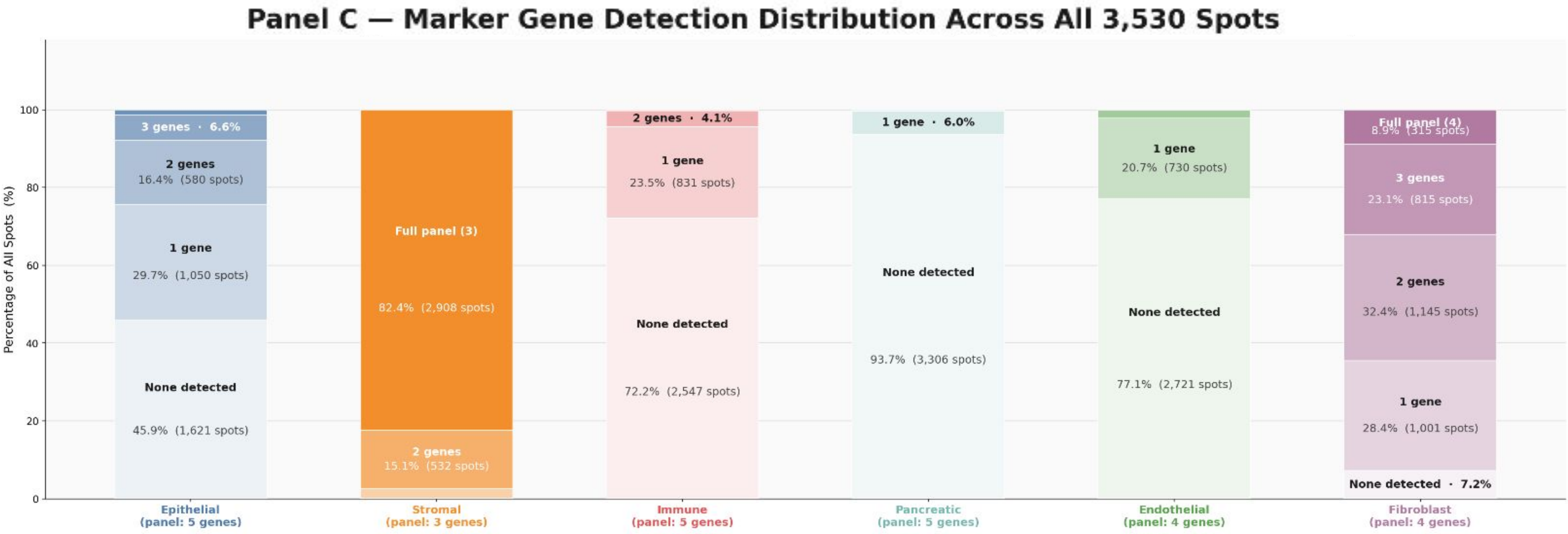
Clinical Validation — Cell Composition Analysis

Table 2 — Tissue-Level Cell Type Summary			
Cell Type	**Mean Expression**	**Patches Expressing**	**Interpretation**
Stromal	**9.23**	2,601 / 3530 (74%)	Dominant cell type — high expression across tissue
Myofibroblast	**3.59**	2,511 / 3530 (71%)	Strong presence — activated fibroblasts
Fibroblast	**1.40**	2,457 / 3530 (70%)	Core component — structural cells
Epithelial	**0.34**	1,927 / 3530 (55%)	Present but lower expression
Macrophage	**0.17**	1,589 / 3530 (45%)	Immune component present
B-Cell	**0.19**	1,038 / 3530 (29%)	Minor immune population
Immune	**0.09**	1,279 / 3530 (36%)	Limited immune infiltration
T-Cell	**0.05**	784 / 3530 (22%)	Sparse T-cell presence
Pancreatic	**0.08**	869 / 3530 (25%)	Tumor cells present
Endothelial	**0.08**	809 / 3530 (23%)	Limited vasculature

Tissue-Level Cell-Type Intrepretation for all spots of IU_PDA_T1 (3530 spots)

Mean expression was calculated by first grouping related cell types into tissue regions (epithelial, stromal, immune, etc.), then computing the average proportion across all spatial spots for each region.

Clinical Validation — Marker Gene Presence Analysis



Each bar is 1 cell type. Stacked segments = how many panel genes were detected per spot.
Labels show % of spots and raw spot counts

Tissue WSI - Patch Locations

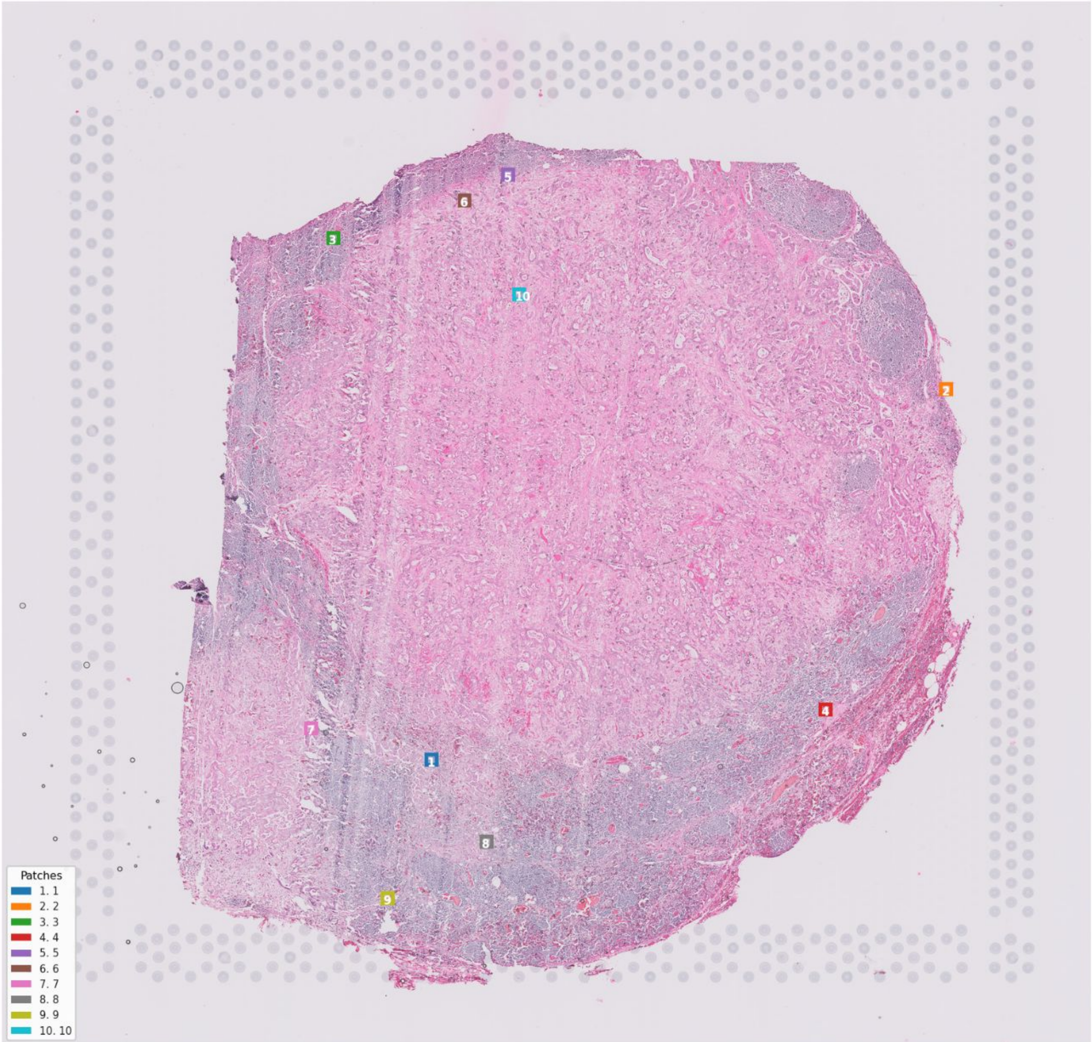


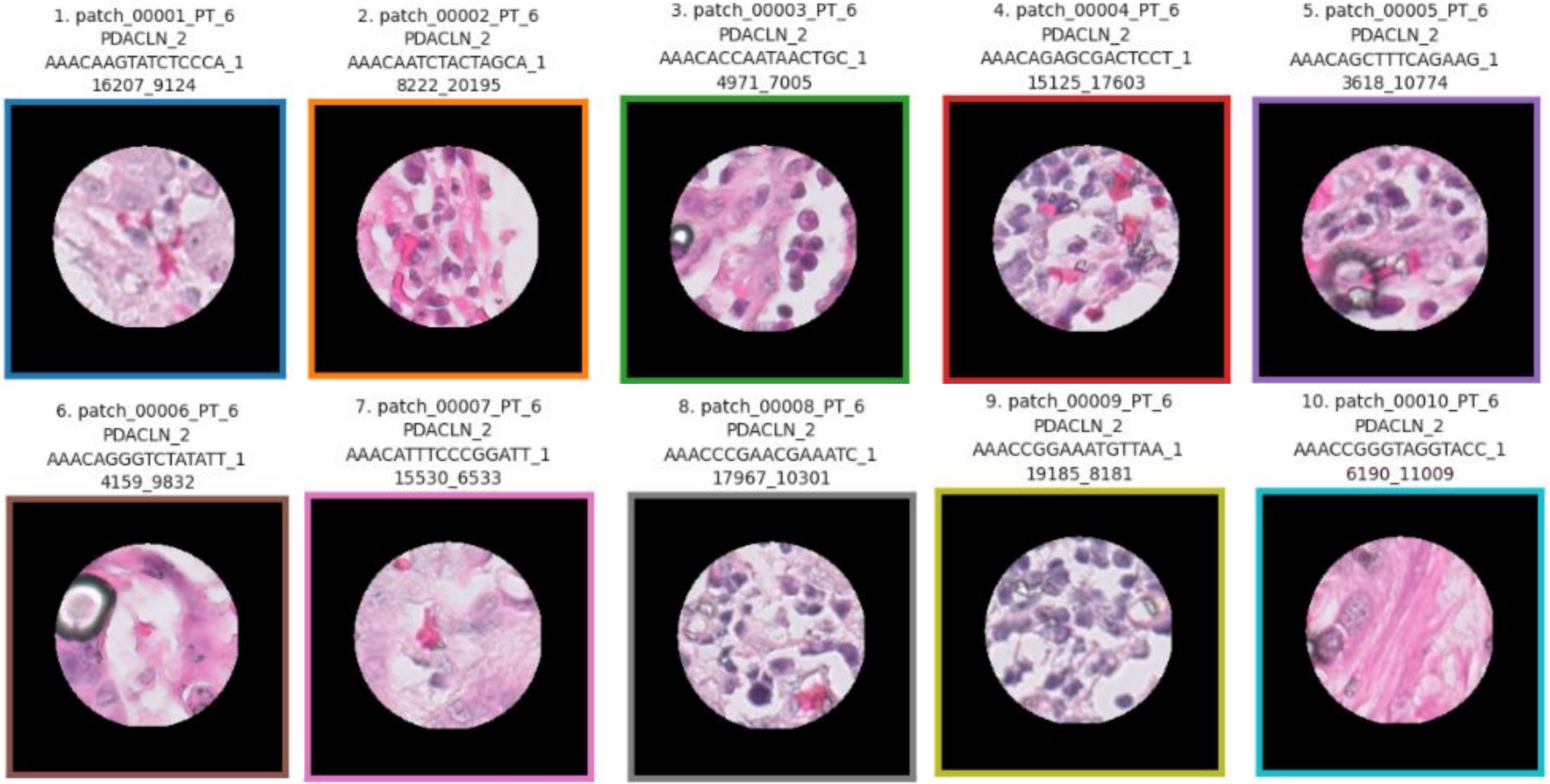
Image:IU_PDA_LNM6

Lymph Node Metastasis Region Image of Patient 6

Patching Process:

- Visium V2, Spot-centered and Spot-sized with black padding patches.
- Patches extracted from .tif files
- First 10 Patches Extracted

First 10 Patches of IU_PDA_LNM6



Clinical Validation — Patient & Patch Metadata | LNM6 (Patient 6)

Table 1A — Patient Clinical Metadata (LNM6)	
Field	**Value**
Patient ID	PT_6
Sample ID	S18-27094_C27
Slide Name	V12M07-119
Area Code	A1
Origin	Lymph node
Age	N/A
Gender	N/A
Ethnicity	N/A
Tumour Location	N/A
AJCC Stage	N/A
Histology	Lymph node
Tumour Grade	N/A
Neoadjuvant Tx	No
Chemo	N/A
R0 Resection	N/A
PNI	N/A
LVI	N/A
Nodes Collected	N/A
Nodes Positive	N/A
Surgery Date	N/A

Clinical Validation — Patient & Patch Metadata | LNM6 (Patient 6)

Table 1B — Visium Scale Information (LNM6)	
Field	**Value**
Mean Pixel Spacing	271.01
Microns / Pixel	0.36899
Pixels / Micron	2.7101
Spot Diameter (px)	149.055
Spot Radius (px)	74.5277
Scale Spot	0.0853534
Scale Fiducial	284.268
Scale Hi-Res	0.0853534
Scale Lo-Res	0.025606

Clinical Validation — Patient & Patch Metadata | LNM6 (Patient 6)

Table 2 — LNM6 Patch / Spot Metadata (top 10 patches · source: patches_metadata.json)							
Patch ID	**Barcode**	**Patient ID**	**Tissue Row**	**Tissue Col**	**Image Row**	**Image Col**	**Patch Filename**
1	PDACLN_2 AAACAAGTATCTCCCA-1	PT_6	50	102	16207	9124	patch_00001_PT_6 PDACLN_2 AAACAAGTATCTCCCA_1 16207_9124
2	PDACLN_2 AAACAATCTACTAGCA-1	PT_6	3	43	8222	20195	patch_00002_PT_6 PDACLN_2 AAACAATCTACTAGCA_1 8222_20195
3	PDACLN_2 AAACACCAATAACTGC-1	PT_6	59	19	4971	7005	patch_00003_PT_6 PDACLN_2 AAACACCAATAACTGC_1 4971_7005
4	PDACLN_2 AAACAGAGCGACTCCT-1	PT_6	14	94	15125	17603	patch_00004_PT_6 PDACLN_2 AAACAGAGCGACTCCT_1 15125_17603
5	PDACLN_2 AAACAGCTTTCAGAAG-1	PT_6	43	9	3618	10774	patch_00005_PT_6 PDACLN_2 AAACAGCTTTCAGAAG_1 3618_10774
6	PDACLN_2 AAACAGGGTCTATATT-1	PT_6	47	13	4159	9832	patch_00006_PT_6 PDACLN_2 AAACAGGGTCTATATT_1 4159_9832
7	PDACLN_2 AAACATTTCCCGGATT-1	PT_6	61	97	15530	6533	patch_00007_PT_6 PDACLN_2 AAACATTTCCCGGATT_1 15530_6533
8	PDACLN_2 AAACCCGAACGAAATC-1	PT_6	45	115	17967	10301	patch_00008_PT_6 PDACLN_2 AAACCCGAACGAAATC_1 17967_10301
9	PDACLN_2 AAACCGGAAATGTTAA-1	PT_6	54	124	19185	8181	patch_00009_PT_6 PDACLN_2 AAACCGGAAATGTTAA_1 19185_8181
10	PDACLN_2 AAACCGGGTAGGTACC-1	PT_6	42	28	6190	11009	patch_00010_PT_6 PDACLN_2 AAACCGGGTAGGTACC_1 6190_11009

All the spot/patch metadata of the 10 patches are shown with patchID and barcode as a unique identifier

Clinical Validation — Gene Expression Analysis

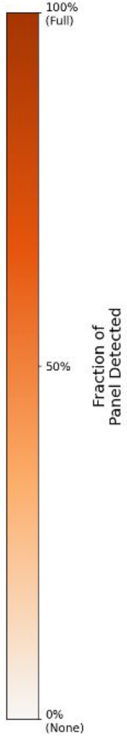
Table 1 — Spot Gene Metadata (first 10 spots)								
Spot ID	Gene Count	Total Expression	Mean Expression	Max Expression	Total Counts	Genes Detected	Norm. Gene Count	% Mitochondrial
1	2592	4389	0.2453	89	4389	2592	0.5906	0
2	1855	3027	0.1692	331	3027	1855	0.6128	0
3	1271	1673	0.0935	18	1673	1271	0.7597	0
4	1466	2099	0.1173	117	2099	1466	0.6984	0
5	4045	7953	0.4445	110	7953	4045	0.5086	0
6	4951	11890	0.6645	152	1.189e+04	4951	0.4164	0
7	3268	6181	0.3454	148	6181	3268	0.5287	0
8	1294	1707	0.0954	32	1707	1294	0.7581	0
9	3096	5566	0.3111	69	5566	3096	0.5562	0
10	4497	9764	0.5457	137	9764	4497	0.4606	0

All the spot gene metadata of the 10 patches are shown with spot ID as a unique identifier
Normalized Gene Count = Genes Detected / Total Counts

Clinical Validation — Marker Gene Presence Analysis

Panel A — Marker Gene Detection Heatmap (10 Spots × 6 Cell Types)

	Epithelial	Stromal	Immune	Pancreatic	Endothelial	Fibroblast
1	3/5 EPCAM KRT8 CDH1	3/3 Collagen Fibronectin Actin	1/5 CD19	0/5 —	0/4 —	3/4 FSP1 VIM PDGFRA
2	1/5 KRT8	3/3 Collagen Fibronectin Actin	1/5 CD45	0/5 —	1/4 PECAM1	3/4 FSP1 VIM PDGFRA
3	2/5 EPCAM KRT8	2/3 Collagen Fibronectin	2/5 CD45 CD19	0/5 —	0/4 —	2/4 FSP1 VIM
4	1/5 KRT8	3/3 Collagen Fibronectin Actin	4/5 CD45 CD3E CD8A CD19	0/5 —	0/4 —	1/4 VIM
5	4/5 EPCAM KRT8 CDH1 TJP1	3/3 Collagen Fibronectin Actin	2/5 CD45 CD3E	0/5 —	1/4 CDH5	3/4 FSP1 VIM THY1
6	4/5 EPCAM KRT8 CDH1 TJP1	3/3 Collagen Fibronectin Actin	2/5 CD3E CD8A	0/5 —	2/4 KDR GATA2	3/4 FSP1 VIM THY1
7	4/5 EPCAM KRT8 CDH1 TJP1	3/3 Collagen Fibronectin Actin	0/5 —	1/5 CTRC	0/4 —	2/4 FSP1 VIM
8	2/5 EPCAM KRT8	3/3 Collagen Fibronectin Actin	0/5 —	0/5 —	0/4 —	1/4 VIM
9	2/5 EPCAM CDH1	3/3 Collagen Fibronectin Actin	2/5 CD45 CD19	0/5 —	2/4 PECAM1 CDH5	2/4 FSP1 VIM
10	3/5 EPCAM KRT8 CDH1	3/3 Collagen Fibronectin Actin	1/5 CD45	0/5 —	0/4 —	3/4 FSP1 VIM THY1

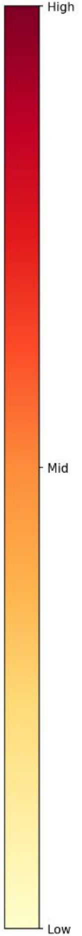


Cell Value = genes found / panel total
Gene names listed = those detected

Clinical Validation — Cell Composition Analysis | LNM6 (Patient 6)

Table 1 — Spot Cell-Type Composition Matrix

Cell Type	Spot 1 PDACLN_2 AAACAAGTATCTCCCA-1	Spot 2 PDACLN_2 AAACAATCTACTAGCA-1	Spot 3 PDACLN_2 AAACACCAATAACTGC-1	Spot 4 PDACLN_2 AAACAGAGCGACTCCT-1	Spot 5 PDACLN_2 AAACAGCTTTCAGAAG-1	Spot 6 PDACLN_2 AAACAGGGTCTATATT-1	Spot 7 PDACLN_2 AAACATTTCCCGGATT-1	Spot 8 PDACLN_2 AAACCCGAACGAAATC-1	Spot 9 PDACLN_2 AAACCGGAAATGTTAA-1	Spot 10 PDACLN_2 AAACCGGGTAGGTACC-1
B cells	0.0755	0.3625	0.2684	0.3248	0.2403	0.0295	0.0225	0.2270	0.3685	0.0072
C1Q-TAM	0.0133	0.0291	0.0072	0.0200	0.0996	0.0100	2.72e-05	0.0232	0.0069	0.0026
CD4+ cells	2.72e-05	0.0802	0.1693	0.1050	0.1313	2.79e-05	2.72e-05	0.1249	0.0384	2.87e-05
CD8-NK cells	2.72e-05	4.14e-05	2.75e-05	2.59e-04	6.55e-05	2.79e-05	0.0033	4.21e-05	2.15e-05	2.87e-05
DCs	0.0521	0.0025	8.24e-05	0.0736	0.0248	2.79e-05	0.0089	0.1218	0.0738	2.87e-05
Endothelial cells	2.72e-05	0.0214	0.0227	0.0196	0.0726	0.0116	0.0109	0.0171	8.90e-04	1.93e-04
FCN1-TAM	2.72e-05	0.0867	0.0986	0.0623	5.15e-05	0.0069	0.0066	0.0129	0.0649	0.0023
Hepatocytes	2.72e-05	4.14e-05	2.75e-05	2.60e-05	5.15e-05	2.79e-05	2.06e-04	3.16e-04	1.75e-05	2.87e-05
iCAF	0.0115	0.1412	0.0447	0.0563	0.0405	2.79e-05	2.72e-05	0.0467	0.0222	2.87e-05
myCAF	0.0748	0.0746	0.0724	0.0061	0.0414	0.0967	0.1744	0.0693	0.0265	0.2379
Normal Epithelial cells	2.72e-05	4.14e-05	2.75e-05	2.60e-05	5.15e-05	2.79e-05	2.72e-05	3.89e-05	1.75e-05	2.87e-05
Proliferative T cells	0.1210	0.1012	0.0977	0.1482	0.1391	0.1296	0.0096	0.1837	0.2847	0.1068
PVL	0.0369	0.0366	0.0332	0.0759	0.1698	0.0441	0.0356	0.0744	0.0458	0.0054
SPP1-TAM	0.0067	4.14e-05	0.0725	0.0052	5.15e-05	2.79e-05	0.0212	3.89e-05	0.0160	0.0084
Tumor Epithelial cells	0.6081	0.0639	0.1130	0.1027	0.0404	0.6715	0.7069	0.0985	0.0514	0.6291



Cell-Type Composition (normalization = Cell type proportion / All cell type proportions of that spot) of 15 Cell Types shown for the 10 Spots

Clinical Validation — Cell Composition Analysis | LNM6 (Patient 6)

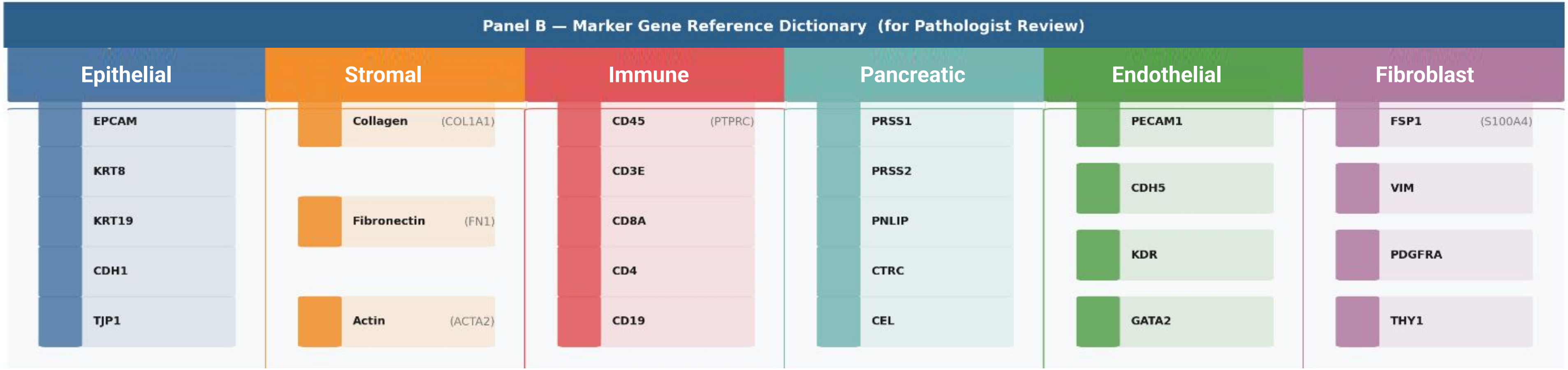
Table 2 — Tissue-Level Cell Type Summary

Cell Type	Mean Expression	Patches Expressing	Interpretation
Epithelial	0.49	3,721 / 3745 (99%)	Dominant cell type — tumor epithelial cells pervasive across tissue
Stromal	0.18	3,745 / 3745 (100%)	Present in all spots — myCAF + iCAF + PVL combined
Immune	0.17	3,736 / 3745 (100%)	Widespread immune infiltration — CD4+, CD8-NK, T cells, DCs
Myofibroblast	0.14	3,665 / 3745 (98%)	Strong presence — activated fibroblasts (myCAF)
T-Cell	0.14	3,732 / 3745 (100%)	Extensive T-cell presence — CD4+, CD8-NK, proliferative T
B-Cell	0.12	3,679 / 3745 (98%)	Significant immune population — B cells broadly expressed
Macrophage	0.04	3,552 / 3745 (95%)	Moderate macrophage presence — C1Q-TAM, FCN1-TAM, SPP1-TAM
Fibroblast	0.01	1,207 / 3745 (32%)	Limited iCAF presence — minority of spots
Endothelial	0.01	1,981 / 3745 (53%)	Moderate vasculature — about half of spots
Pancreatic	0.00	53 / 3745 (1%)	Near absent — consistent with lymph node metastasis site

Tissue-Level Cell-Type Intrepretation for all spots of IU_PDA_LNM6 (3745 spots)

Mean expression was calculated by first grouping related cell types into tissue regions (epithelial, stromal, immune, etc.), then computing the average proportion across all spatial spots for each region.

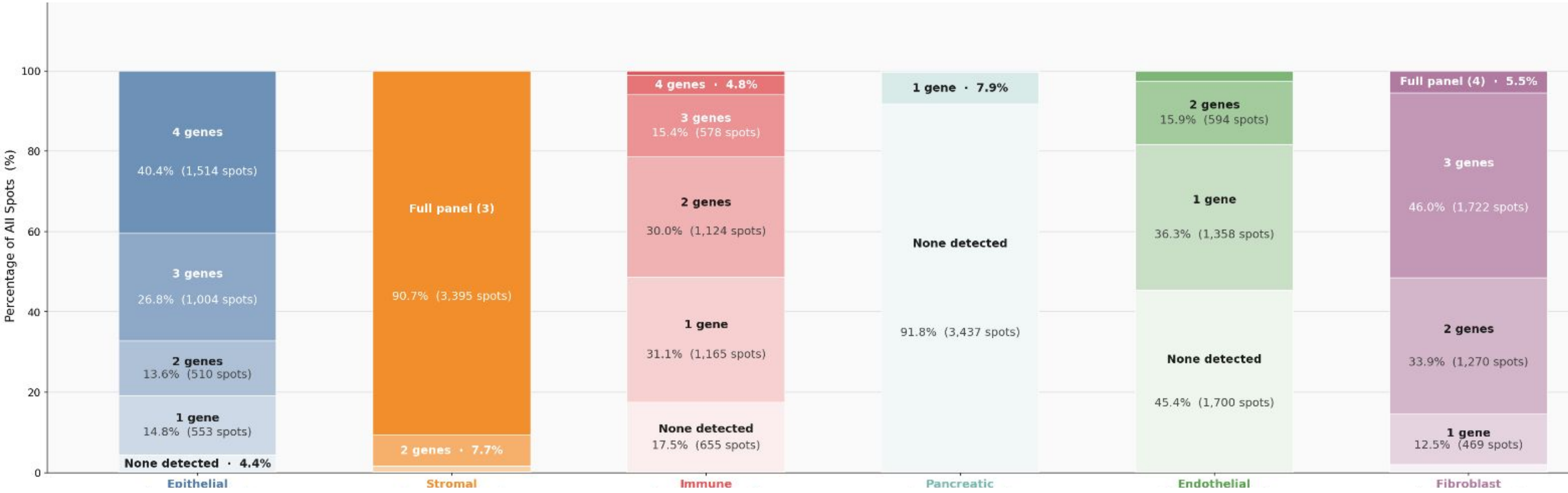
Clinical Validation — Marker Gene Presence Analysis



This is the Marker Gene Reference Dictionary where key marker genes have been used to identify the tissue regions in IU_PDA_LNM6

Clinical Validation — Marker Gene Presence Analysis

Panel C – Marker Gene Detection Distribution Across All 3,745 Spots



Each bar is 1 cell type. Stacked segments = how many panel genes were detected per spot.
Labels show % of spots and raw spot counts

Zenodo Spatial Transcriptomics First Approach Implementation

Dataset Overview

Tissue Sample	n_spots
IU_PDA_T1	3530
IU_PDA_T3	4354
IU_PDA_T4	3621
IU_PDA_T11	2777
IU_PDA_HM11	3931
IU_PDA_HM13	2182

Pair

Training done on 6 samples:

- 4 Primary Tumour Samples
- 2 Hepatic Metastasis Samples

Zenodo Dataset contained for each sample:

- .TIF Images
- ST Data (count matrix)
- scRNA Data
- Clinical metadata

Zenodo Spatial Transcriptomics Project Updates

Project Objectives & Motivation

This project has designed objectives in progressing towards identifying specific morphological features in primary pancreatic tumors that predict liver metastasis. They are divided into 3 Phases (This meeting covers the first 2):

Phase A - Knowledge Distillation & Embedding Augmentation

What?: The model must understand the relationship between morphological features and its spatial transcriptomics (gene expression & cell composition) for each spot respectively

How?: By developing a transformer-based architecture that learns to represent the transcriptomic profile within the visual embedding space.

Result: A model capable of ST-free inference, that is taking an H&E slide and generating an "augmented embedding" that inherently contains the predicted biological state of those cells

Zenodo Spatial Transcriptomics Project Updates

Project Objectives & Motivation

Phase B - Identifying the "Metastatic Signature"

What?: The model must distinguish between tumor regions that are "prone to metastasize" (specifically to the liver) and those that are not.

How?: Use the augmented embeddings of model from Phase A to identify which specific "spots" or "features" in the primary tumor most closely resemble the confirmed metastatic cells in the liver on the basis of similar gene expression.

Result: Develop a scoring system to tell the clinician which specific spots in this primary tumor are already "metastasized" in their behavior and the likelihood of liver-specific metastasis based on current tumor morphology.

Zenodo Spatial Transcriptomics Project Updates

Phase A: Pre-requirements

In understanding the relationship between morphology and spatial transcriptomics, 2 experiments are designed to track the modality dependency:

- 3 Modality Architecture: Vision, Gene Expression and Cell Composition
- 2 Modality Architecture: Vision, Gene Expression

This experiment was design with the purpose of checking the dependency of scRNA data, which is crucial data needed in extracting cell deconvolution results. As the objectives of this phase was to eliminate the need of ST data into using morphological-only model, scRNA is another kind of data which would monitored into making **100% “Morphological-Only”!**

We started with 3 Modality Architecture first

Zenodo Spatial Transcriptomics Project Updates

Phase A: 3-Modality Contrastive Learning CLIP-style Architecture

Vision Projector (Pv): A small MLP that maps z to a joint space S.

Gene Projector (Pg): An MLP that maps the Gene Expression vector (g) to the same space S.

Cell Projector (Pc): An MLP that map Cell composition vector to the same space S

- **Training Objective:** Use **InfoNCE Loss** (Contrastive Loss).
  - Maximize similarity between a patch and its *actual* ST spot.
  - Minimize similarity between a patch and *other* ST spots

Zenodo Spatial Transcriptomics Project Updates

Phase A: 3-Modality Contrastive Learning CLIP-style Architecture

Embedding Extraction:

- **Vision** — UNI2-h (1536-dim), H-Optimus-1 (1536-dim), CONCH v1 (512-dim) patch embeddings, one 224×224 circular crop per Visium spot
- **Gene** — scVI (50-dim) latent embeddings, trained on 20,395 spots, 17,893 genes, negative binomial likelihood
- **Cell** — RCTD (15-dim) cell type composition vectors, 15 cell types deconvolved per spot

All embeddings saved as per-spot .pt tensors in mirrored per-sample directories.

TriModal Bridge Architecture

Three independent MLP projectors mapping each modality into a shared 256-dim latent space:

- Pv (Vision): 1536 → 768 → 384 → 256 (3 layers, deepest)
- Pg (Gene): 50 → 512 → 256 (2 layers)
- Pc (Cell): 15 → 128 → 64 → 256 (3 layers, narrowest hidden dims)

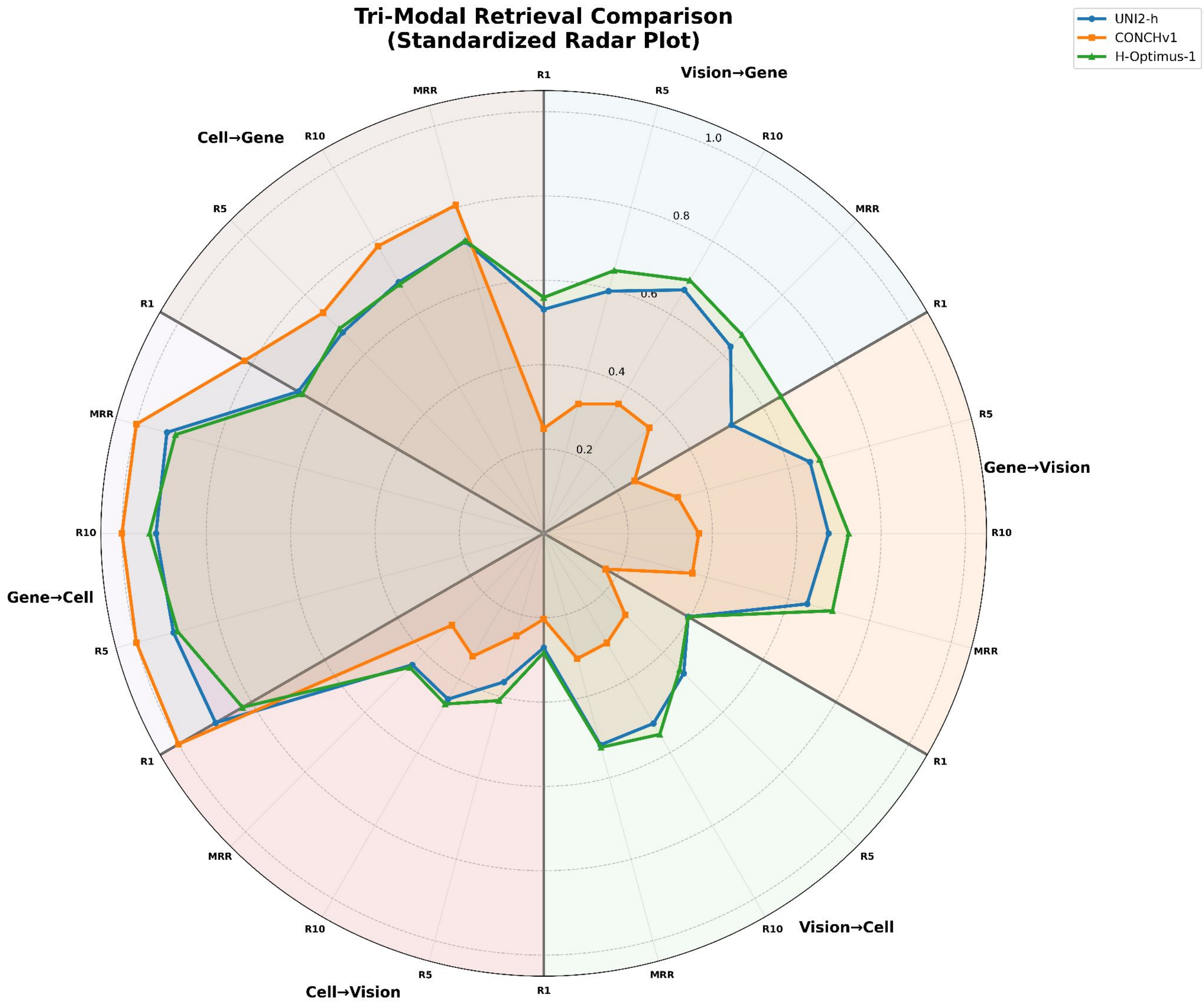
Zenodo Spatial Transcriptomics Project Updates

Phase A: 3-Modality Contrastive Learning CLIP-style Architecture

Trained with **weighted InfoNCE loss** across three modality pairs (v↔g, v↔c, g↔c), vision-upweighted (×2.0). Learnable per-pair temperatures clamped to [0.01, 0.5]. 5-fold cross-validation, effective batch size 256 via gradient accumulation.

The dataset distribution in 5-fold cross validation is:

Split	Proportion	Exact Spots
Test split	15%	3,060
Train + Validation Split	85%	17,335



Tri-Modal Retrieval Comparison
(Standardized Radar Plot)
UNI2-h
CONCHv1
H-Optimus-1
Vision→Gene
Gene→Vision
Vision→Cell
Cell→Vision
Gene→Cell
Cell→Gene
R1
R5
R10
MRR
0.2
0.4
0.6
0.8
1.0

Zenodo Spatial Transcriptomics Project Updates

Key Observations:

1. **g→c is the best direction**! This makes biological sense as gene expression and cell composition are directly related in PDAC tissue. scVI and RCTD are both derived from the same count matrix, so their embeddings share latent structure even after separate projection.
2. **Vision↔gene pairs (v→g, g→v) outperform vision↔cell pairs (v→c, c→v)**. This suggests the UNI2-h, CONCHv1 and H-Optimus-1 visual features correlate more with transcriptome variation than with cell type composition. Histological texture encodes gene programs more directly than it encodes cell proportions, which makes sense as cell type deconvolution is a harder inference problem from morphology alone.

Zenodo Spatial Transcriptomics Project Updates

Phase B - Identifying the "Metastatic Signature":

Since this phase involves our model learning to differentiate between PT and HM samples on the basic of their gene expression patterns, 4 modules have been designed for its architecture:

1. **Module 1 - Alignment**

InfoNCE Loss maps each of the 256-dim modality onto the same unit sphere, real embeddings of spots with ST are pulled together while different embeddings are pushed away.

The output embeddings from Phase A is averaged together and normalized to create a fused embedding (256-dim) for Phase B.

It then passes by a small MLP, trained by Supervised Contrastive Loss (SupCon). **For every batch of spots, positive pairs are defined as: one PT spot and one HM spot whose scVI latents have cosine similarity > 0.7.** That threshold means "these two spots express genes in similar proportions are molecularly related regardless of which cohort they came from." Negative pairs are all other PT-HM combinations in the batch.

Zenodo Spatial Transcriptomics Project Updates

Phase B - Identifying the "Metastatic Signature":

2. **Module 2 - Metastatic Direction**

Two centroids are computed in the 128d aligned space, **Δ_meta** is a vector in 128d space pointing from the average PT spot toward the average HM spot. It's the direction of metastatic transformation in embedding space.

This is the cumulative effect of gene expression changes, cell type shifts, and morphological differences. A PT spot with a high direction score has already partially traversed this axis. It hasn't “metastasised”, but its biology is pointing in that direction.

3. **Module 3 - MLP**

Trained by regression against the Module 2 direction scores as soft labels — so it's learning to predict the same quantity, but as a learned non-linear function rather than a geometric projection.

Loss is MSE between predicted score and direction score. Training uses the full aligned embeddings from Module 1.

Zenodo Spatial Transcriptomics Project Updates

Phase B - Identifying the "Metastatic Signature":

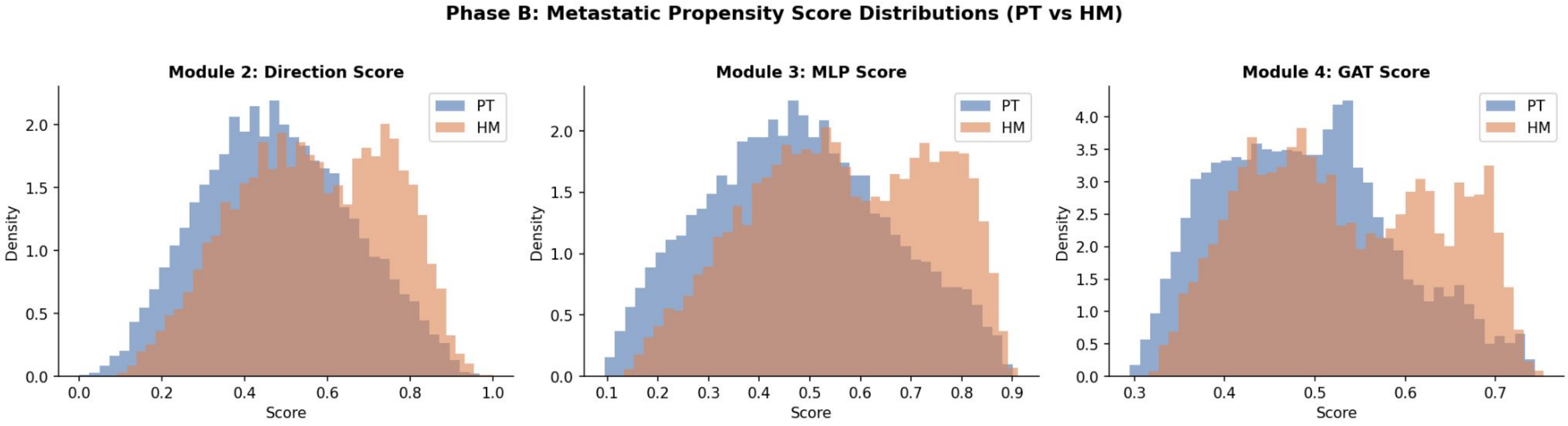
4. **Module 4 - The GAT Score**

Builds a spatial graph over the Visium slide: each spot is a node, connected to its 6 nearest neighbours by tissue coordinate proximity (standard Visium hex grid). Each node's feature vector is its 128d aligned embedding concatenated with its Module 3 score.

A Graph Attention Network learns attention weights for each edge — how much should each neighbour influence the central spot's final score. The output is a smoothed score per spot that incorporates neighbourhood context.

Zenodo Spatial Transcriptomics Project Updates

Phase B - Identifying the "Metastatic Signature":



This result is for UNI2-h, scVI, RCTD model. This results shows that PT and HM are largely overlapping with each other. This tells that subpopulation of PT spots already occupies the metastatic region of embedding space.

Zenodo Spatial Transcriptomics Project Updates

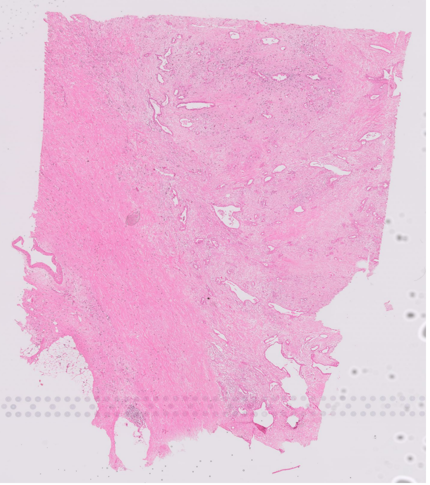
Phase B - Identifying the "Metastatic Signature":

Key Observations:

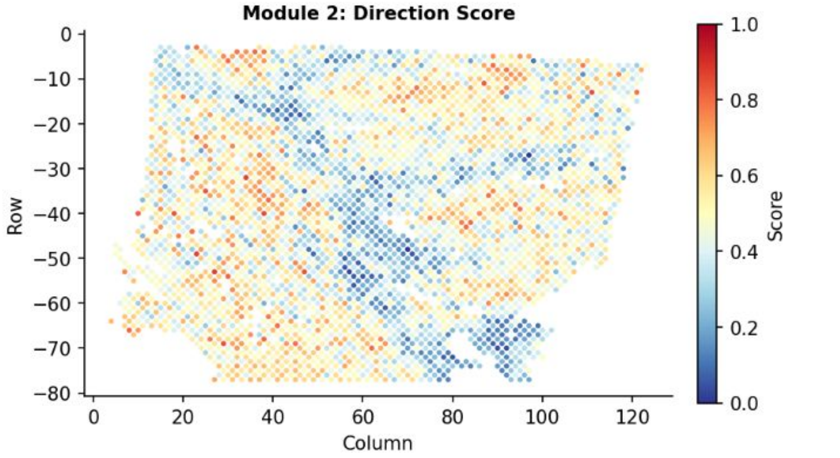
1. **The metastatic signal exists and is linear:** Modules 2 and 3 produced nearly identical separation (both $\Delta \approx 0.088$). When a geometric projection and a learned neural network give the same answer, it means the biology separating PT from HM in your embedding space organises along one dominant linear axis.
2. **The metastatic signal is spatially diffuse in PT tissue:** The GAT score halved the cohort separation from 0.088 down to 0.043. This is the most biologically informative observation of the entire phase. When spatial smoothing collapses your signal, it means high-scoring PT spots are scattered across the tissue rather than clustered.

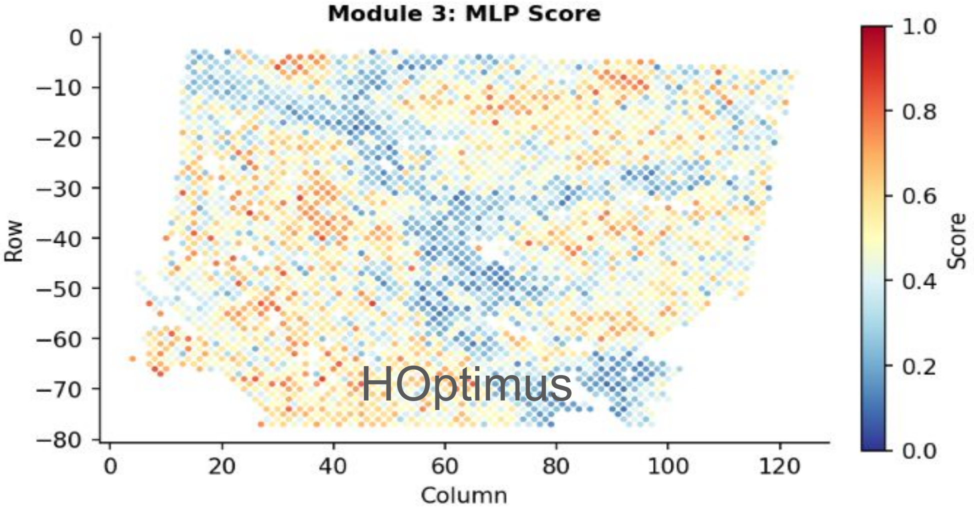
IU_PDA_T1

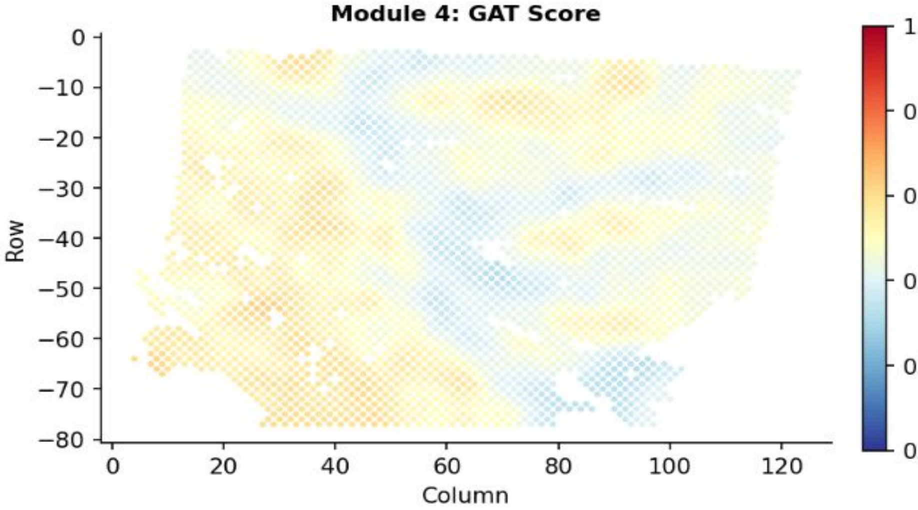
3530 Spots



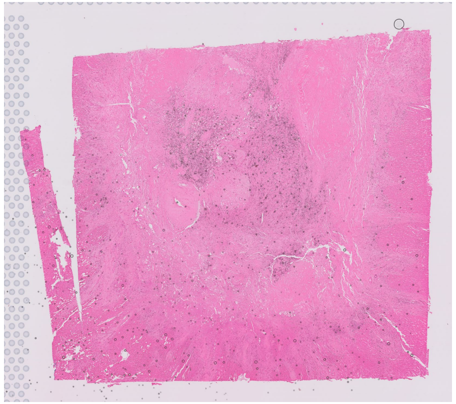
HOptimus



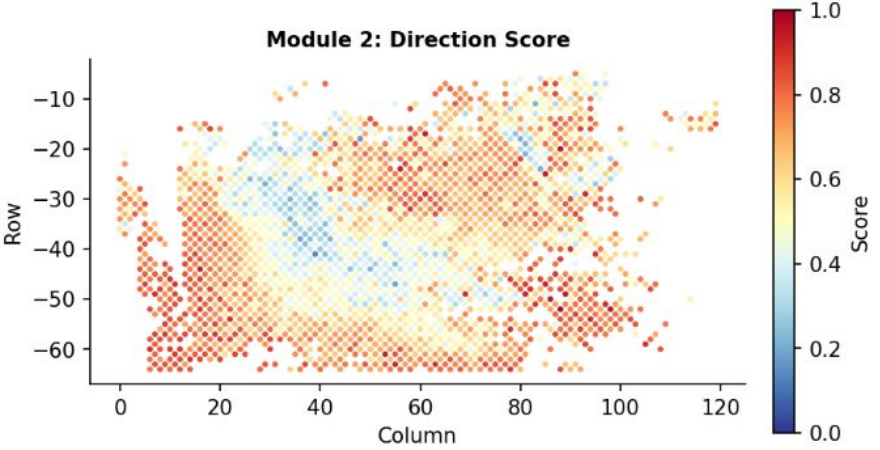


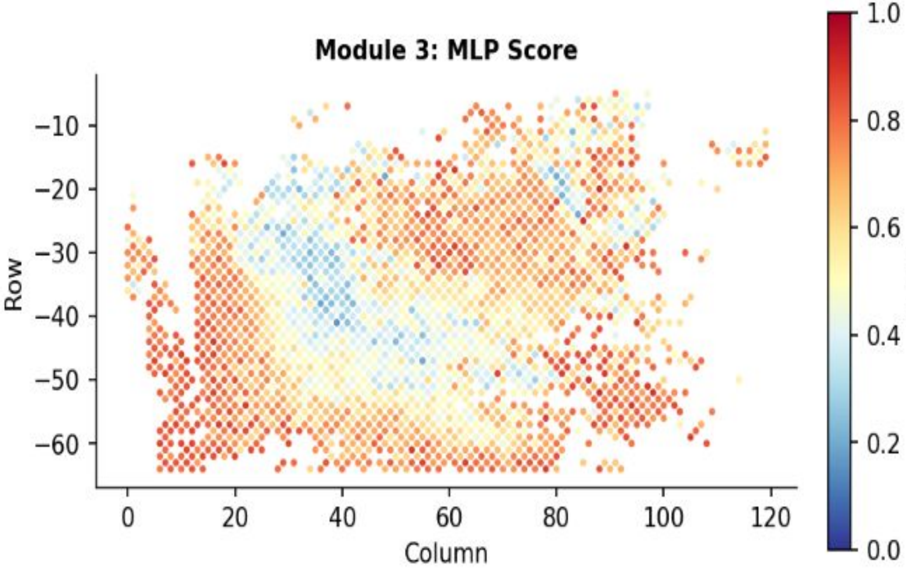


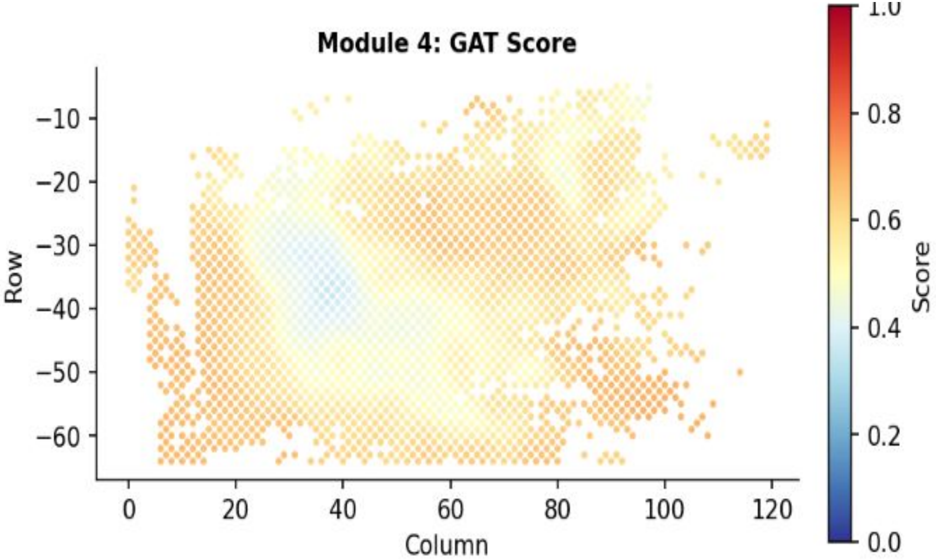
IU_PDA_HM13
2182 Spots



HOptimus



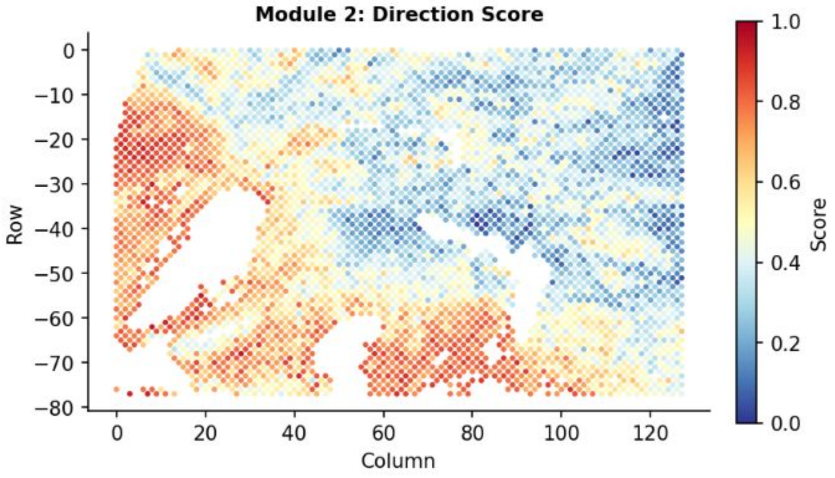


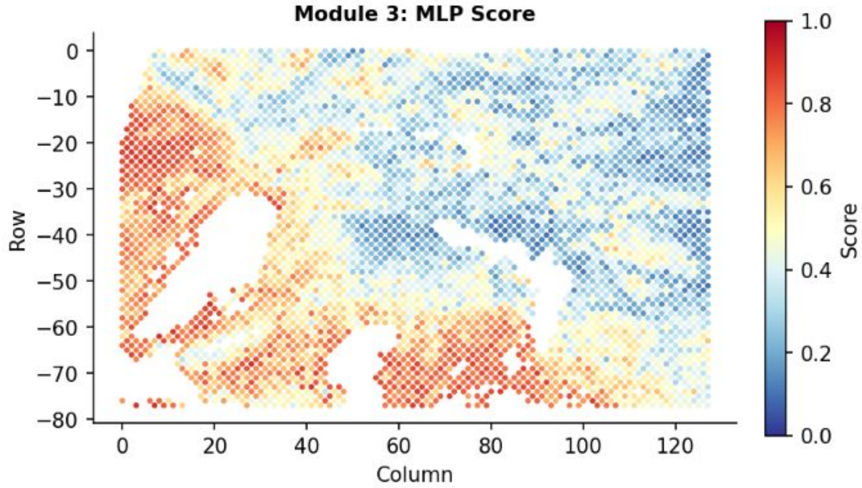


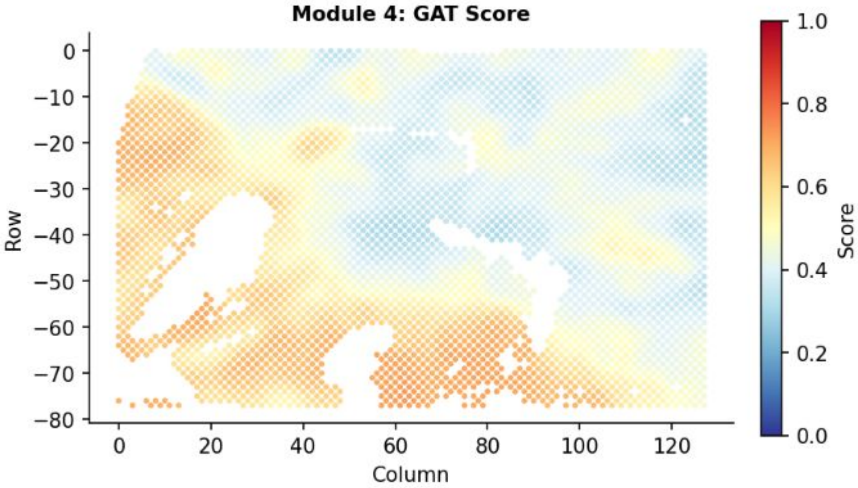
IU_PDA_T3

4354 Spots

H-Optimus-1, scVI, RCTD



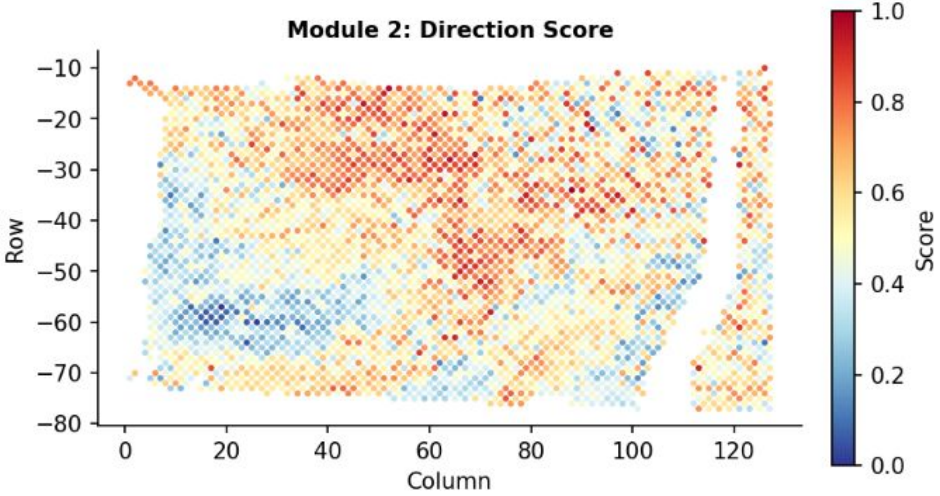


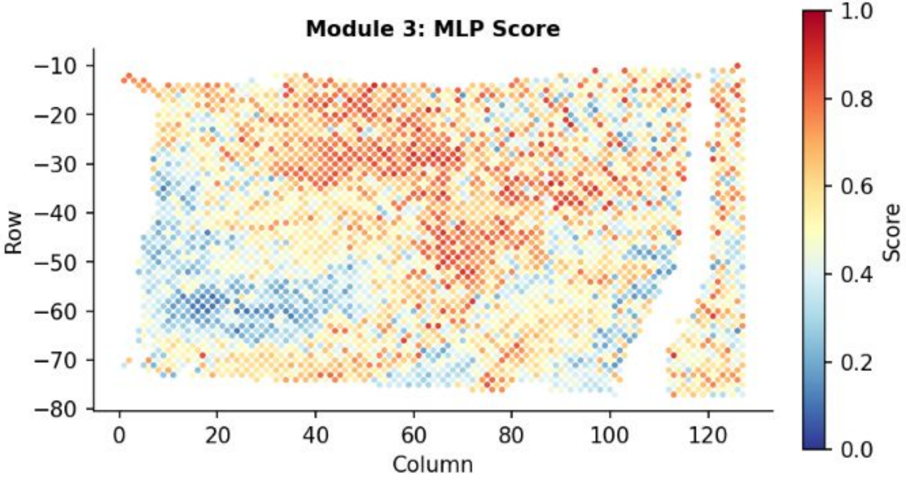


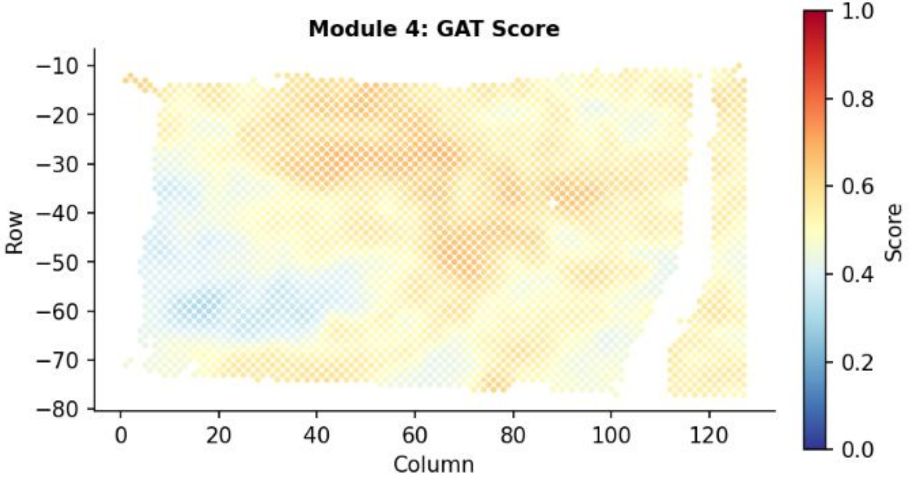
IU_PDA_T4

3621 Spots

H-Optimus-1, scVI, RCTD



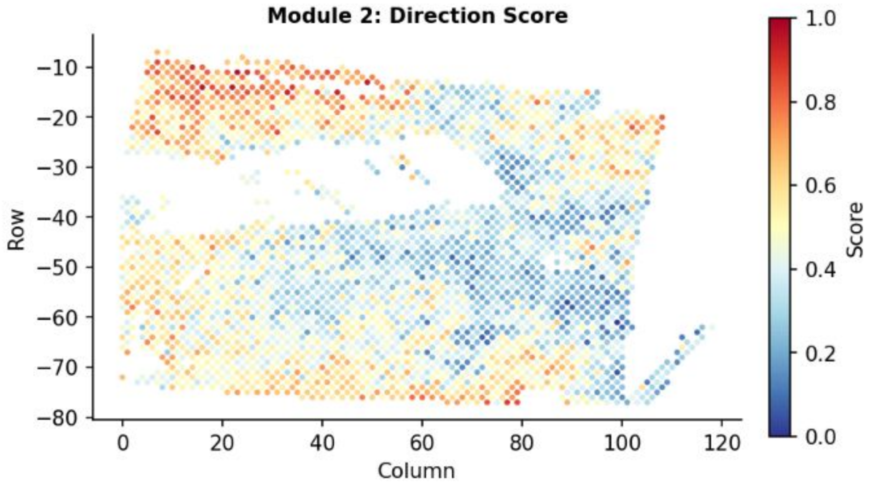


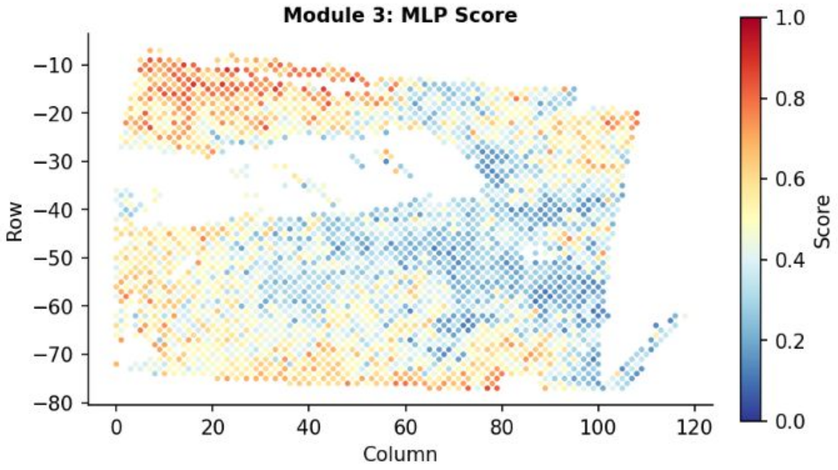


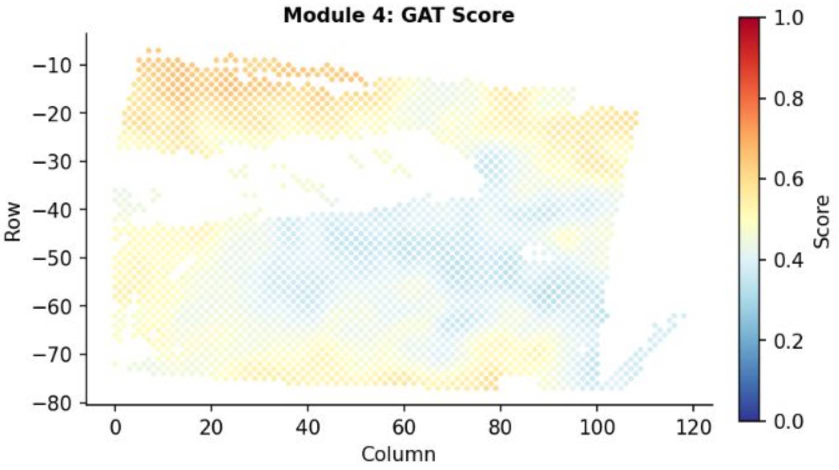
IU_PDA_T11

2777 Spots

H-Optimus-1, scVI, RCTD







IU_PDA_HM11

3931 Spots

H-Optimus-1, scVI, RCTD